[Geochemistry, Geophysics, Geosystems]

Supporting Information for

Constraining Solid Dynamics, Interface Rheology, and Slab Hydration in the Hikurangi Subduction Zone Using 3D Fully Dynamic Models

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Introduction

This supplementary document provides additional information regarding the numerical methods and data post processing for “Constraining Solid Dynamics, Interface Rheology, and Slab Hydration in the Hikurangi Subduction Zone Using 3D Fully Dynamic Models”. In detail, we (1) outline the approximation for the Peierls creep deformation mechanism used in this study, (2) provide additional details regarding the plate velocity misfit calculation, (3) present an estimate of modeled trench retreat, (4) highlight the sensitivity of rheology to strain rate where Peierls creep is active, (5) illustrate the effect of crustal rheology on intra-slab strain rates, and (6) illustrate the effect of the addition of the Alpine fault weak-zone on large-scale flow patterns.

Peierls Creep Approximation

We use an approximate version the Peierls creep mechanism (equation 7) for computational efficiency. This approximate version is expressed by:

$$\eta_p = A \cdot B \cdot C$$

$$A = \frac{0.5\gamma\sigma_p}{(A_p(\gamma\sigma_p)^n)^{\frac{1}{s+n}}}$$

$$B = \exp\left(\frac{E+PV}{RT} \cdot p \cdot q \cdot (1-\gamma^p)^{q-1} \cdot \gamma^p\right)$$

$$C = \dot{\epsilon}^{\frac{1}{s+n}-1}$$

$$s = \frac{pq(E+PV)}{RT} \cdot (1-\gamma^p)^{q-1}\gamma^p$$

where γ is a fitting parameter set to 0.17 and other parameters are defined in Table 1 in the main publication. A value of 0.17 is chosen for γ because this is the most appropriate value for the high stresses within the subducting plate.

Calculating Velocity Misfit

We determine the misfit in convergence rate between the ASPECT models (Figure S1a) and the MORVEL velocities (Figure S1b) at every point on the modeled Pacific plate east of the slab. Therefore, we do not calculate the misfit in convergence north of 26˚S where we terminate the slab for computational efficiency (Figure S1c). The largest misfit occurs close to the trench (Figure S1c), where the modeled convergence transitions from dominantly east-west convergence to include a radial (downward) component. The misfit peaks at the northern extent of the modeled slab.

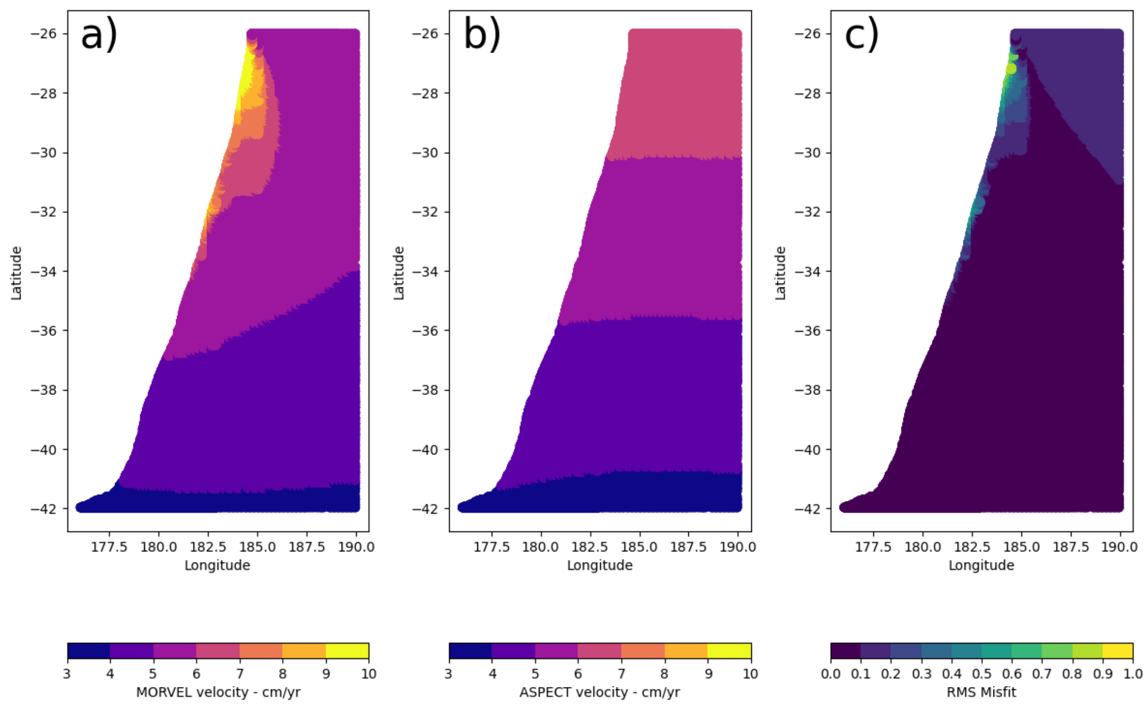


Figure S1. Surface velocities of the Pacific plate from (a) the best fitting ASPECT model (interface viscosity = 5e20 Pa s, dry rheology), and (b) the MORVEL model (DeMets et al., 2010). The RMS misfit between the ASPECT velocities and the MORVEL velocities is shown in panel c).

Modeled Trench Motion

To quantify the migration of the trench, we extract the location of the modeled trench and draw a parallel curve on either side. Next, we query the magnitude of the velocity at each point on both of these trench-parallel curves and determine the trench motion using

$$v_{trench} = v_{Pac} - v_{Aus},$$

where, v_{trench}, v_{Pac}, and v_{Aus} is the velocity of the trench, the velocity of the Pacific plate, and the velocity of the Australian plate, respectively. Therefore, when the velocity is negative, the trench retreats, and when the velocity is positive, the trench advances.

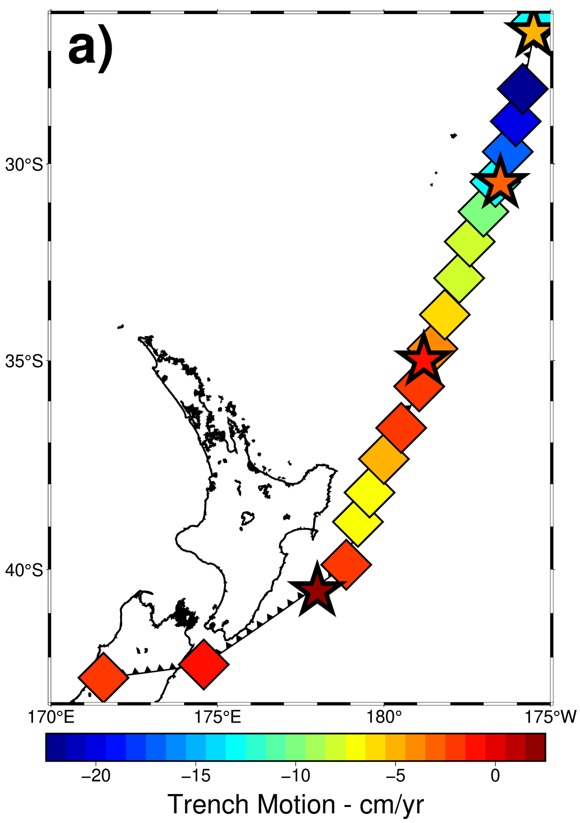


Figure S2. Estimated trench motion for an interface viscosity of 5e20 Pa s (coloured diamonds) at each point along the modeled trench as defined by the Geodynamic WorldBuilder for (a) the base plate configuration and (b) a model with the Alpine Fault and Bird 2003 plate boundary model extending from the southern Alpine Fault to the Southern model boundary. Estimates of trench motion from Schellart and Spakman, 2012 are also provided (coloured stars). The black sawtooth line marks the location of the modeled trench, and the solid black contour shows the coastline of New Zealand. Negative (positive) trench motion corresponds to trench retreat (advance)

Nonlinear Rheological Feedbacks

A somewhat counterintuitive outcome of our model results is that the dry rheological models predict higher convergence rates than the wet rheological models. In the absence of low temperature plasticity (Peierls creep), and under the same pressure, temperature, and strain rate, diffusion and dislocation creep predict lower viscosities for a wet rheology compared to a dry one. For a dry rheology to behave more weakly than a wet rheology, the dry rheology must experience a significantly higher strain rate, on the order of 10x. However, when low-temperature plasticity (Peierls creep) is introduced, even a slightly higher strain rate, as low as a 20% increase, can cause a dry rheology to exhibit a lower viscosity than a wet rheology (Figure S3).

In models with a wet crust, the weaker crust results in a more diffuse zone deformation zone along the plate boundary, lowering the maximum local strain rates compared to the stronger dry rheology. Consequently, the Peierls mechanism results in a slightly higher viscosity along the interface contributing to slower convergence rates. Additionally, in the wet crustal model, the crust at depth has lower viscosity than in the dry model (Figure S4), resulting in greater viscous decoupling between the subducting crust and the subducting mantle lithosphere. This weakens the transfer of slab pull force up-dip toward the trench, further reducing convergence rates.

Another unexpected result is that extending the weak zone south into the Alpine Fault decreases convergence rates relative to our base plate boundary configuration. This can be explained by deformation accommodated along the Alpine Fault, which includes a segment sub-parallel to the Pacific–Australian convergence direction. This geometry encourages more southward Pacific Plate motion in the model with the Alpine Fault, reducing strain rates along the Hikurangi margin to the north. Additionally, the Alpine Fault influences the movement of the Australian Plate, further altering the stress state along the entire Hikurangi–Kermadec Trench (Figure S5).

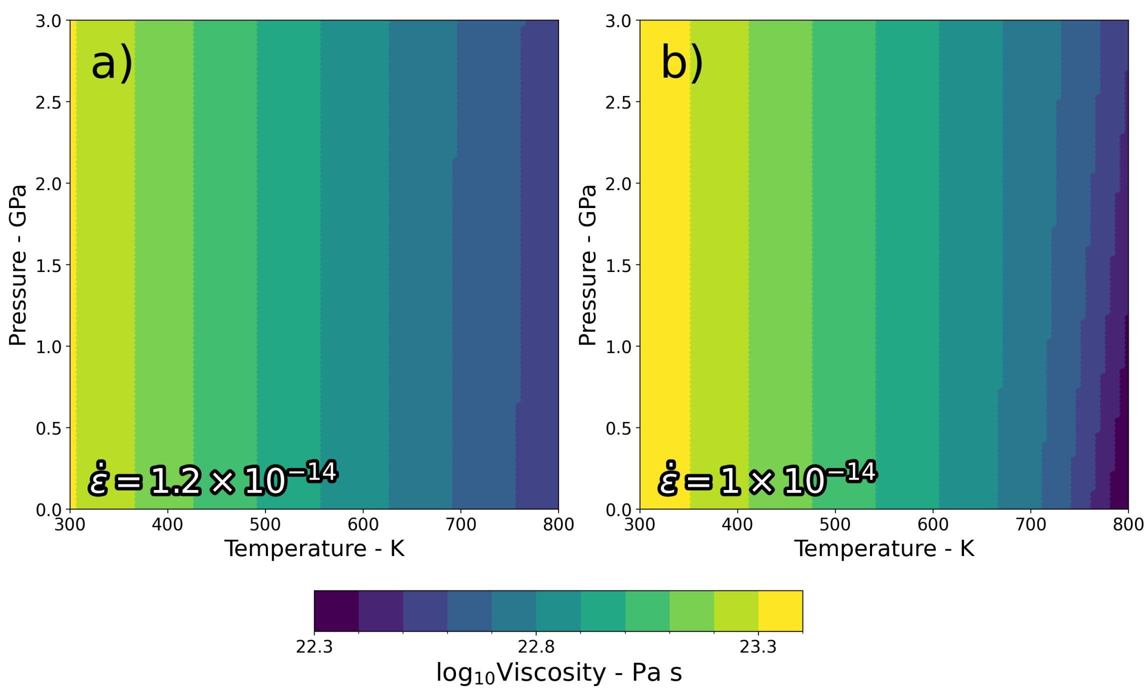
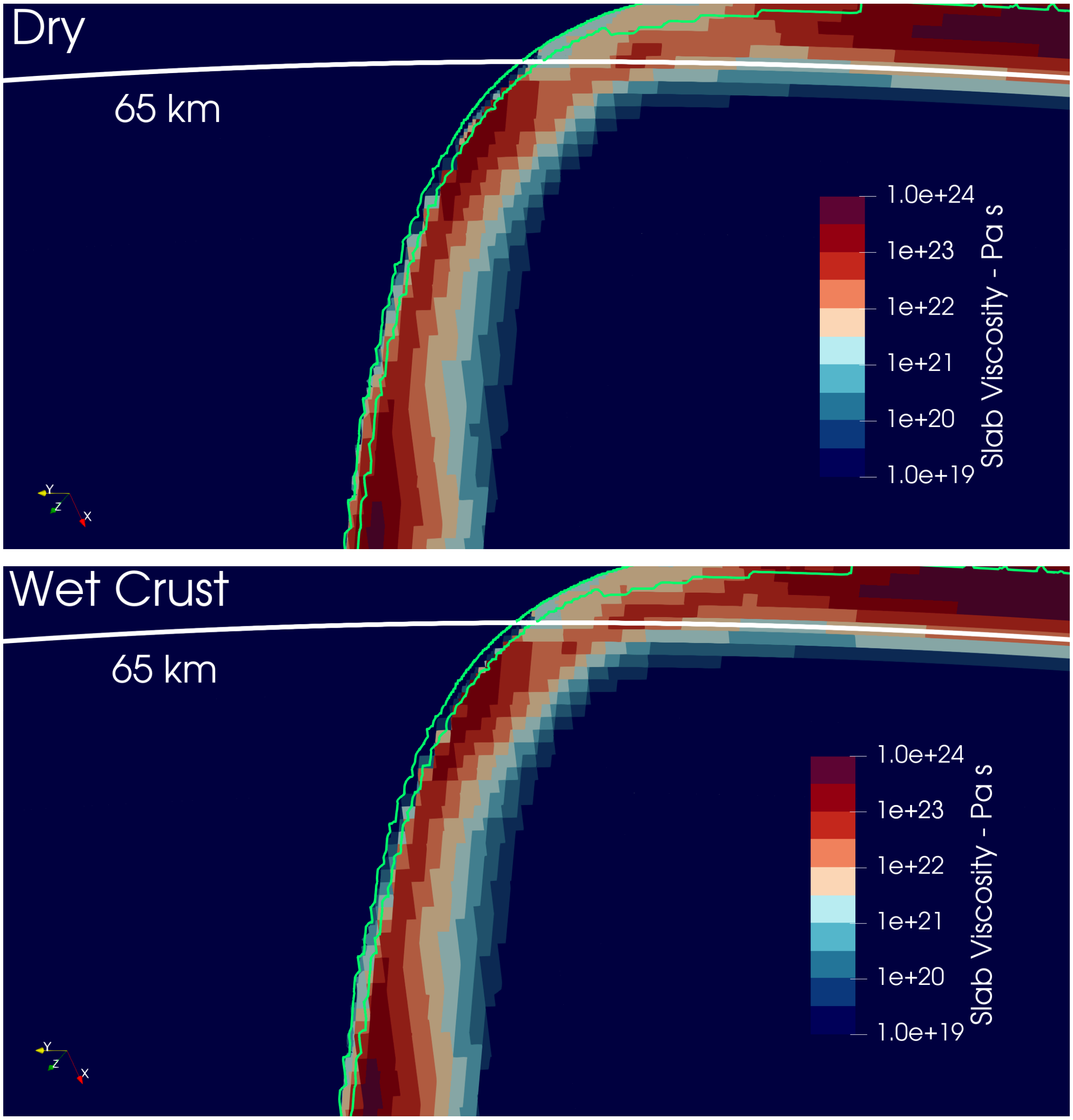


Figure S3. A simple test which demonstrates the sensitivity of the modeled viscosity to the strain rates due to the inclusion of the Peierls creep mechanism over a pressure and temperature range representative of the plate interface. The viscosity distribution described in the main text for a (a) dry rheology and (b) a homogeneous wet rheology illustrates that slightly increasing the strain rate (1.2e-14 compared to 1e-14) produces



lower viscosities in the dry rheology due to the highly nonlinear dependence of the Peierls creep mechanism on strain rate.

Figure S4. Comparison of the viscosity within the subducting crust for models containing an interface viscosity of 5e20 Pa s, the base plate boundary configuration, and a dry rheology (top) or a homogeneously hydrated crust (bottom). The white line delineates the depth of the basalt-eclogite reaction. Viscosities within the wet crust model below the basalt-eclogite transition are lower than the dry rheology model, which promotes mechanical decoupling within the slab and may affect the magnitude of slab pull transmitted through the plate. The green contour shows the location of the subducting oceanic crust.

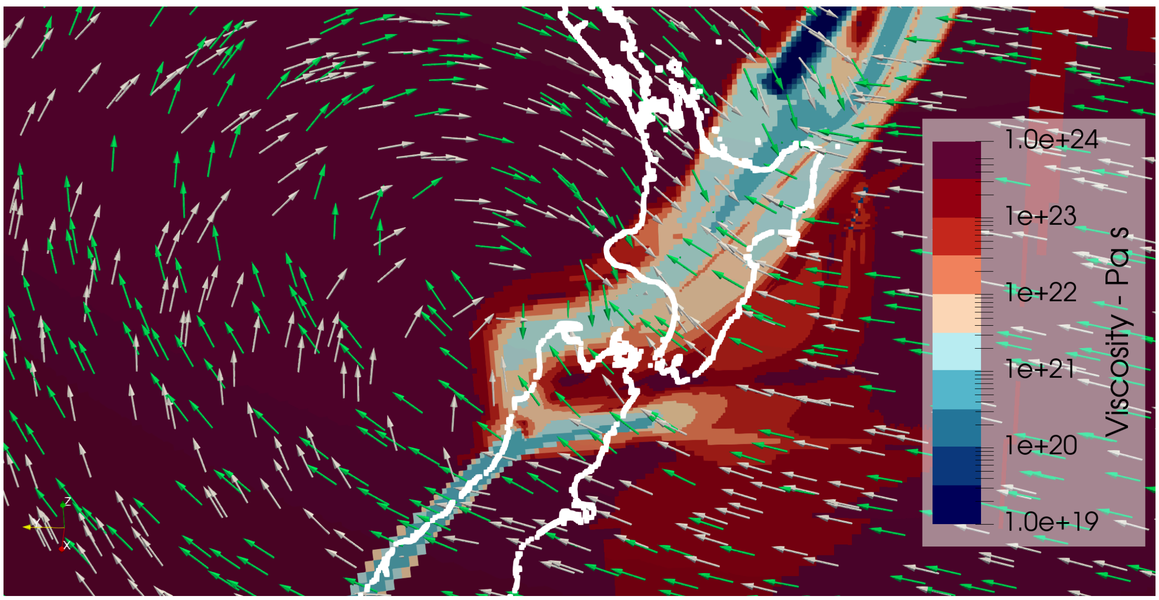


Figure S5. Depth cross section at 10 km depth coloured by the modeled viscosity from the model which prescribes a 5e20 Pa s interface viscosity, a dry rheology, and includes the Alpine Fault. The coastline of New Zealand is shown as a white contour. Vectors show the direction of flow for the base plate configuration (green vectors), and base plate configuration with the added Alpine Fault weak-zone (white vectors).

Bibliography

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